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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2019 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Monday 19th August 2019

General Instructions

- Reading time — 5 minutes
- Writing time — 2 hours
- Write using black pen.
- NESA-approved calculators and templates may be used.

Total — 70 Marks

- All questions may be attempted.

Section I — 10 Marks

- Questions 1–10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eleven.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference Sheet
- Candidature — 143 boys

Examiner

SDP

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following is an even function?

- (A) $y = x$
- (B) $y = 2^x$
- (C) $y = (x - 2)^4$
- (D) $y = \sqrt{5 - x^2}$

QUESTION TWO

Which of the following is equal to $\int \frac{dx}{4 + x^2}$?

- (A) $\cos^{-1} 2x + C$
- (B) $2 \sin^{-1} x + C$
- (C) $\frac{1}{2} \tan^{-1} \frac{x}{2} + C$
- (D) $\log_e(1 + x^2) + C$

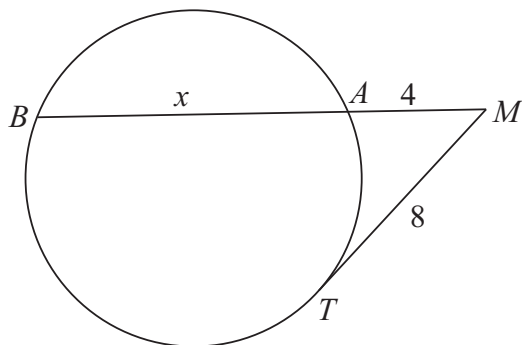
QUESTION THREE

Which of the following is equal to $\frac{100!}{98! \times 2!}$?

- (A) $100 \times 99 \times 98$
- (B) 100×99
- (C) 50×99
- (D) $50 \times 49\frac{1}{2}$

Examination continues next page ...

QUESTION FOUR



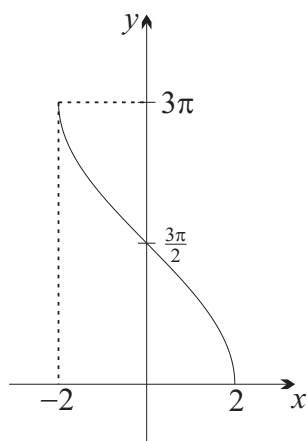
Suppose TM is a tangent to a circle at T , while MB is a secant intersecting the circle at A and B . Given that $TM = 8$, $AB = x$ and $MA = 4$, what is the value of x ?

- (A) $2\sqrt{17} - 2$
- (B) 12
- (C) 14
- (D) 16

QUESTION FIVE

Which of the following is the primitive of $\cos^2 x$?

- (A) $x + \frac{1}{2} \cos 2x + C$
- (B) $x - \frac{1}{2} \cos 2x + C$
- (C) $\frac{1}{2}x + \frac{1}{4} \sin 2x + C$
- (D) $\frac{1}{2}x - \frac{1}{4} \sin 2x + C$

QUESTION SIX

Which equation is best represented by the graph above?

- (A) $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$
- (B) $y = 6 \sin^{-1} \left(\frac{x}{2} \right)$
- (C) $y = \frac{3}{2} \cos^{-1} (2x)$
- (D) $y = 2 \sin^{-1} (x)$

QUESTION SEVEN

Which of the following polynomials are divisible by $x + 1$?

- (I) $x^{2019} - 1$ (II) $x^{2019} + 1$ (III) $x^{2020} - 1$ (IV) $x^{2020} + 1$
- (A) (I) and (III) only
- (B) (II) and (IV) only
- (C) (II) and (III) only
- (D) (I) and (IV) only

QUESTION EIGHT

Which of the following equations is true, given that $\ddot{x} = 2x(3x - 1)$?

- (A) $v = 2x^3 - x^2 + C$
- (B) $v^2 = 2x^3 - x^2 + C$
- (C) $v = x^2(x^3 - x) + C$
- (D) $v^2 = 4x^3 - 2x^2 + C$

QUESTION NINE

What is the derivative of $y = \sqrt{1 + \sqrt{x}}$?

- (A) $\frac{1}{2\sqrt{1 + \sqrt{x}}}$
- (B) $\frac{1}{\sqrt{x}\sqrt{1 + \sqrt{x}}}$
- (C) $\frac{1}{2\sqrt{x}\sqrt{1 + \sqrt{x}}}$
- (D) $\frac{1}{4\sqrt{x}\sqrt{1 + \sqrt{x}}}$

QUESTION TEN

What is the value of $\tan(\alpha + \beta)$ if $\tan \alpha + \tan \beta + 4 = \cot \alpha + \cot \beta = 10$?

- (A) $\frac{3}{5}$
- (B) $\frac{5}{3}$
- (C) 6
- (D) 15

————— End of Section I —————

Examination continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.**Marks**

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x}$. 1

(b) Find the domain of the function $y = 4\sin^{-1}(2x - 3)$. 1

(c) The equation $x^3 + 6x^2 - 2x + 4 = 0$ has roots α , β and γ . Find the value of:

(i) $\alpha + \beta + \gamma$ and $\alpha\beta + \beta\gamma + \gamma\alpha$, 1

(ii) $\alpha^2 + \beta^2 + \gamma^2$. 2

(d) The volume of water in a lake increases over time. The flow of water into the lake is given by the flow rate $\frac{dV}{dt} = 120(3 - \sin 2t)$ where V is the volume of water in the lake in cubic metres at time t in days.

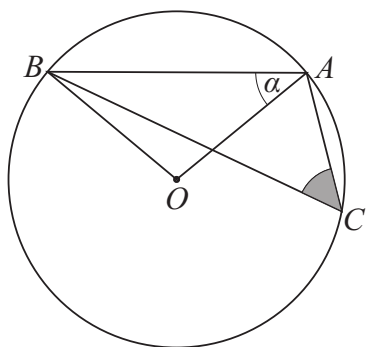
(i) What is the maximum flow rate of water? 1

(ii) Given that the lake has initial volume of 5000 m^3 find V in terms of t . 2

(e) Differentiate $y = \tan^{-1}(\log_e x)$. Give your answer in simplest form. 2

(f) Evaluate $\int_e^{e^3} \frac{dx}{2x \ln x}$ using the substitution $u = \ln x$. Give your answer in exact form. 3

(g)



In the diagram above O is the centre of the circle. Points A , B and C all lie on the circumference of the circle. If $\angle OAB = \alpha$ find the size of $\angle ACB$. Give reasons for your answer. 2

Examination continues next page ...

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) Solve $\frac{3}{x} < 2$. 2

(b) Prove that $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$. 1

(c) A particle P is moving in a straight line with its motion given by $\ddot{x} = -16x$ where x is the displacement of P from the origin O . Initially P is 3 metres on the right of O and is moving towards O with velocity $4\sqrt{3}$ m/s.

(i) Show that the speed of the particle is given by $4\sqrt{12 - x^2}$ m/s. 2

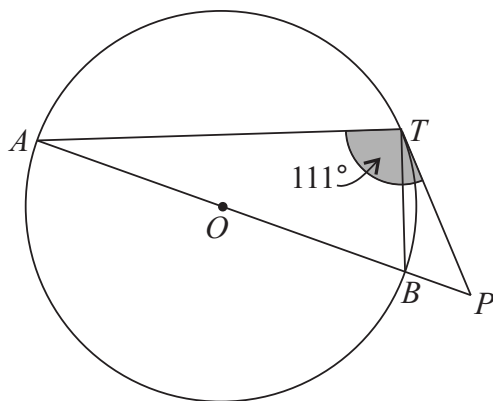
(ii) Verify that $x = A \sin(4t + \alpha)$ satisfies $\ddot{x} = -16x$ for all values of the constants A and α . 1

(d) Let $f(x) = \frac{1}{5}x - \log_e x$.

(i) Show that $f(x)$ has a root between $x = 1$ and $x = e$. 1

(ii) Taking $x_1 = 1.5$ as an initial approximation, use Newton's method once to obtain x_2 , a better approximation of the root. Write down the value of x_2 correct to two decimal places. 2

(e)



In the diagram above AB is a diameter of the circle, TP is a tangent at point T , O is the centre of the circle and $\angle ATP = 111^\circ$. Find $\angle BAT$ giving reasons. 2

(f) A chord PQ joins the points $P(2p, p^2)$ and $Q(2q, q^2)$ on the parabola $x^2 = 4y$. The chord PQ passes through the point $A(0, 2)$.

(i) Derive the equation of the chord PQ . 1

(ii) Find the coordinates of M , the midpoint of PQ . 1

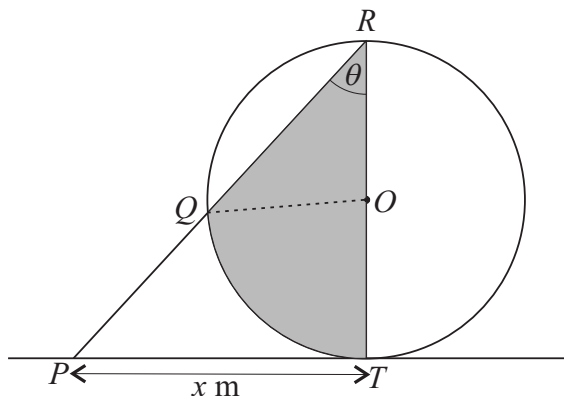
(iii) Show that $pq = -2$. 1

(iv) Hence find the equation of the locus of M as P and Q vary. 1

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) Using $t = \tan \frac{\theta}{2}$ show that $\sin \theta + \cos \theta = \frac{1}{4}$ can be written as $5t^2 - 8t - 3 = 0$. **2**
- (ii) Hence solve $\sin \theta + \cos \theta = \frac{1}{4}$ for $-\pi < \theta < \pi$. Give your answer correct to one decimal place. **2**

(b)



The above diagram shows a circle centre O , with radius 1 metre. The line PT of length x m is a tangent to the circle at T and RT is a diameter. The line PR cuts the circle at Q . Let $A \text{ m}^2$ be the area of the shaded region and let $\angle ORQ = \theta$ in radians. The point P is moving away from T at a constant speed of 16 m/s.

- (i) Express $\tan \theta$ in terms of x and find θ when $x = \frac{2}{\sqrt{3}}$. **1**
- (ii) Find $\frac{d\theta}{dt}$ when $x = \frac{2}{\sqrt{3}}$. **2**
- (iii) Show that $A = \theta + \frac{1}{2} \sin 2\theta$. **2**
- (iv) Find $\frac{dA}{dt}$ when $x = \frac{2}{\sqrt{3}}$. **2**
- (c) (i) Let t_r be the coefficient of x^r in the expansion of $(a + bx)^n$. Show that: **2**
- $$\frac{t_{r+1}}{t_r} = \frac{n-r}{r+1} \times \frac{b}{a}.$$
- (ii) Hence, or otherwise, find the coefficients of the two consecutive terms that have equal coefficients in the expansion of $(2 + 3x)^{14}$. **2**

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) The polynomial $P(x)$ is divided by $(x + 2)(x - 5)$. Find the remainder given that $P(-2) = 6$ and $P(5) = -1$. **2**

- (b) A particle is projected from the origin on level ground with speed 15 m/s at an angle α to the horizontal. Let the acceleration due to gravity be $g = 10 \text{ m/s}^2$.

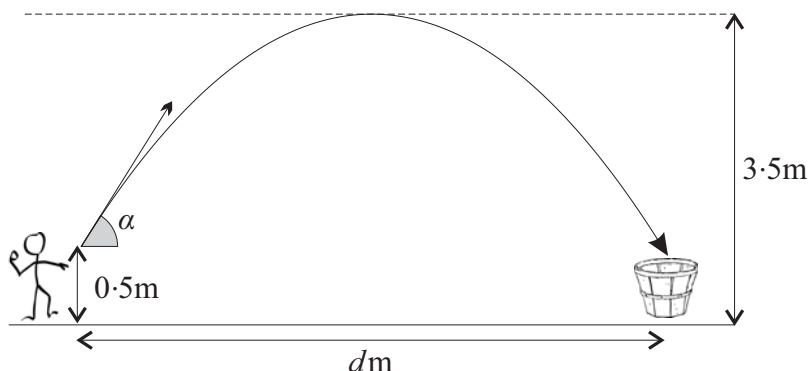
- (i) Derive the equations for x and y , the horizontal and vertical displacement of the particle respectively in terms of t . **1**

- (ii) Show that the maximum height reached by the particle h metres is given by **2**

$$h = \frac{45}{4} \sin^2 \alpha.$$

- (iii) Show that the particle returns to the initial height at $x = \frac{45}{2} \sin 2\alpha$. **2**

- (iv) Sophie throws a paper ball into the centre of a bin across a room. The paper ball is projected from a point 0.5 m above the floor and the top of the bin is also 0.5 m above the floor. The ceiling height is 3.5 m above the floor.



The paper ball is thrown with a velocity 15 m/s at an angle of α . Assuming no air resistance, show that the maximum separation d metres that Sophie and the bin can have and still get the paper ball into the bin is $d = 6\sqrt{11} \text{ m}$. **2**

- (c) Use mathematical induction to show that for any integer $n \geq 0$, **3**

$$\sum_{r=0}^n \frac{1}{2^r} \tan \left(\frac{x}{2^r} \right) = \frac{1}{2^n} \cot \left(\frac{x}{2^n} \right) - 2 \cot (2x),$$

where $0 < x < \frac{\pi}{4}$.

- (d) If the roots of the quadratic equation $8x^2 - 5x + a = 0$ are $\sin \theta$ and $\cos 2\theta$ for some angle θ , find the possible values of a . **3**

————— End of Section II —————

END OF EXAMINATION

Extension 1 Trial Solutions 2019

1. (D) ✓

(A) $f(-x) = -x$ not even

(B) $f(-x) = 2^{-x}$ not even

(C) $f(-x) = (-x-2)^4 = (x-2)^4$ not even

(D) $f(-x) = \sqrt{5 - (-x)^2} = \sqrt{5 - x^2}$ so even

2. (C) ✓

$$\begin{aligned}\int \frac{1}{4+x^2} dx &= \int \frac{1}{2^2+x^2} dx \\ &= \frac{1}{2} \tan^{-1} \frac{x}{2} + C\end{aligned}$$

3. (C) ✓

$$\begin{aligned}\frac{100!}{98! \times 2!} &= \frac{100 \times 99 \times 98!}{98! \times 2} \\ &= \frac{100 \times 99}{2} \\ &= 50 \times 99\end{aligned}$$

4. (B) ✓

$$\begin{aligned}AM \times MB &= TM^2 \quad (\text{tangent and secant}) \\ 4 \times (x+4) &= 8^2 \\ 4x+16 &= 64 \\ 4x &= 48 \\ x &= 12\end{aligned}$$

5. (C) ✓

$$\begin{aligned}\int \cos^2 x \, dx &= \frac{1}{2} \int (\cos 2x + 1) \, dx \quad (\text{from double angle formula}) \\ &= \frac{1}{2} \left(\frac{1}{2} \sin 2x + x \right) + C \\ &= \frac{1}{2} x + \frac{1}{4} \sin 2x + C\end{aligned}$$

6. (A) ✓

$\cos^{-1} x$ domain is $-1 \leq x \leq 1$.

$\cos^{-1} x$ range is $0 \leq y \leq \pi$.

so this graph must be $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$.

7. (C)

✓

If divisible by $(x + 1)$ then $P(-1) = 0$

$$\begin{aligned}
 \text{(I)} \quad P(-1) &= ((-1)^{2019} - 1) \\
 &= -1 - 1 \\
 &= -2 \quad \text{so not divisible by } x + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(II)} \quad P(-1) &= ((-1)^{2019} + 1) \\
 &= -1 + 1 \\
 &= 0 \quad \text{so divisible by } x + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(III)} \quad P(-1) &= ((-1)^{2020} - 1) \\
 &= 1 - 1 \\
 &= 0 \quad \text{so divisible by } x + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{(IV)} \quad P(-1) &= ((-1)^{2020} + 1) \\
 &= 1 + 1 \\
 &= 2 \quad \text{so not divisible by } x + 1
 \end{aligned}$$

8. (D)

✓

$$\begin{aligned}
 \frac{d}{dx}(\tfrac{1}{2}v^2) &= 6x^2 - 2x \\
 \tfrac{1}{2}v^2 &= 2x^3 - x^2 + \tfrac{1}{2}C \\
 v^2 &= 4x^3 - 2x^2 + C
 \end{aligned}$$

9. (D)

✓

$$\begin{aligned}
 y &= \left(1 + x^{\frac{1}{2}}\right)^{\frac{1}{2}} \\
 \frac{dy}{dx} &= \tfrac{1}{2} \times \tfrac{1}{2}x^{-\frac{1}{2}} \times \left(1 + x^{\frac{1}{2}}\right)^{-\frac{1}{2}} \\
 &= \frac{1}{2 \times 2 \times \sqrt{x} \times \sqrt{1 + \sqrt{x}}} \\
 &= \frac{1}{4\sqrt{x}\sqrt{1 + \sqrt{x}}}
 \end{aligned}$$

10. (D)

✓

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan \alpha + \tan \beta + 4 = 10$$

$$\tan \alpha + \tan \beta = 6$$

$$\cot \alpha + \cot \beta = 10$$

$$\frac{1}{\tan \alpha} + \frac{1}{\tan \beta} = 10$$

$$\frac{\tan \alpha + \tan \beta}{\tan \alpha \tan \beta} = 10$$

$$\frac{6}{\tan \alpha \tan \beta} = 10$$

$$\tan \alpha \tan \beta = \frac{3}{5}$$

$$\begin{aligned}\tan(\alpha + \beta) &= \frac{6}{1 - \frac{3}{5}} \\ &= 15\end{aligned}$$

11. (a)

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sin 2x}{3x} &= \frac{2}{3} \times \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \\
 &= \frac{2}{3} \times 1 \\
 &= \frac{2}{3} \quad \checkmark
 \end{aligned}$$

(b) Domain of $4\sin^{-1}x$ is $-1 \leq x \leq 1$
 Domain of $4\sin^{-1}(2x-3)$ is:

$$\begin{aligned}
 -1 &\leq 2x-3 \leq 1 \\
 2 &\leq 2x \leq 4 \\
 1 &\leq x \leq 2 \quad \checkmark
 \end{aligned}$$

(c) i. $\alpha + \beta + \gamma = -6$ and $\alpha\beta + \beta\gamma + \gamma\alpha = -2$ $\checkmark$ (both)
 ii.

$$\begin{aligned}
 \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \quad \checkmark \\
 &= (-6)^2 - 2 \times -2 \\
 &= 40 \quad \checkmark
 \end{aligned}$$

(d) i. Maximum flow rate occurs when $\sin 2t = -1$
 So when $\sin 2t = -1$,

$$\begin{aligned}
 \frac{dV}{dt} &= 120(3 - (-1)) \\
 \frac{dV}{dt} &= 480 \text{ m}^3/\text{day} \quad \checkmark
 \end{aligned}$$

ii.

$$\begin{aligned}
 \frac{dV}{dt} &= 120(3 - \sin 2t) \\
 V &= 120\left(3t + \frac{1}{2}\cos 2t\right) + C \quad \checkmark \\
 V &= 5000 \text{ when } t = 0 \\
 5000 &= 120 \times \frac{1}{2} + C \\
 C &= 4940 \\
 V &= 120\left(3t + \frac{1}{2}\cos 2t\right) + 4940 \quad \checkmark \text{ (or similar)}
 \end{aligned}$$

(e)

$$\begin{aligned}
 f(x) &= \tan^{-1}(\log_e x) \\
 f'(x) &= \frac{1}{x} \times \frac{1}{1 + (\log_e x)^2} \quad \checkmark \\
 &= \frac{1}{x(1 + (\log_e x)^2)} \quad \checkmark \text{ (or similar)}
 \end{aligned}$$

(f)

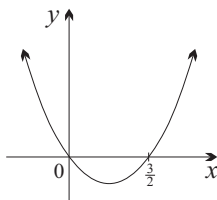
$$\begin{aligned}
 \int_e^{e^3} \frac{1}{2x \ln x} dx &= \int_1^3 \frac{1}{2u} du && \checkmark && u = \ln x && x = e^3, u = 3 \\
 &= \frac{1}{2} [\ln u]_1^3 && \checkmark && \frac{du}{dx} = \frac{1}{x} && x = e, u = 1 \\
 &= \frac{1}{2} [\ln 3 - \ln 1] && && du = \frac{1}{x} dx \\
 &= \frac{1}{2} \ln 3 && \checkmark
 \end{aligned}$$

- (g) $OB = OA$ (radii) so $\triangle OAB$ is isosceles
 $\angle OBA = \alpha$ (equal base angles in isosceles triangle)
 $\angle AOB = 180^\circ - 2\alpha$ (angles in a triangle) $\checkmark$

$$\begin{aligned}
 \angle ACB &= \frac{1}{2} \times (180^\circ - 2\alpha) \text{ (angle at circumference is half angle at centre)} \\
 &= 90^\circ - \alpha && \checkmark \text{ (must have reasons for both marks)}
 \end{aligned}$$

12. (a)

$$\begin{aligned}
 \frac{3}{x} &< 2 \text{ multiply both sides by } x^2 \\
 3x &< 2x^2 && \checkmark \\
 0 &< x(2x - 3) \\
 \text{from graph, } x &< 0 \text{ or } x > \frac{3}{2} && \checkmark
 \end{aligned}$$



(b)

$$\begin{aligned}
 RHS &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \times \frac{dv}{dx} \\
 &= v \frac{dv}{dx} \\
 &= \frac{dx}{dt} \times \frac{dv}{dx} \\
 &= \frac{dv}{dt} \\
 &= \ddot{x} && \checkmark \text{ (show that question)} \\
 &= LHS
 \end{aligned}$$

(c) i.

$$\ddot{x} = -16x$$

$$\frac{d}{dx} \left(\frac{1}{2}v^2 \right) = -16x$$

$$\frac{1}{2}v^2 = \frac{-16}{2}x^2 + \frac{1}{2}C$$

$$v^2 = -16x^2 + C$$

initially, $x = 3$ and $v = 4\sqrt{3}$

$$(4\sqrt{3})^2 = -16 \times 3^2 + C$$

$$48 = -144 + C$$

$$C = 192 \quad \checkmark$$

$$\text{so } v^2 = 192 - 16x^2$$

$$|v| = \sqrt{192 - 16x^2} \quad (\text{as speed is positive})$$

$$= 4\sqrt{12 - x^2} \quad \checkmark \text{ (show that question)}$$

ii.

$$x = A \sin(4t + \alpha)$$

$$v = 4A \cos(4t + \alpha)$$

$$\ddot{x} = -16A \sin(4t + \alpha)$$

$$= -16x \quad \checkmark$$

$$(d) \quad i. \quad f(1) = \frac{1}{5} - \log_e 1 = \frac{1}{5}$$

$$f(e) = \frac{1}{5} \times e - \log_e e \approx -0.46$$

Therefore as $f(1)$ is positive and $f(e)$ is negative, it has a root between 1 and e as $f(x)$ is continuous. $\checkmark$ (both values found)

$$ii. \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f'(x) = \frac{1}{5} - \frac{1}{x} \quad \checkmark$$

$$x_2 = 1.5 - \frac{\frac{1}{5} \times 1.5 - \log_e 1.5}{\frac{1}{5} - \frac{1}{1.5}}$$

$$\approx 1.27 \quad \checkmark$$

$$(e) \quad \angle ATB = 90^\circ \text{ (Thales Theorem)}$$

$$\angle BTP = 111^\circ - 90^\circ = 21^\circ \text{ (adjacent angles)} \quad \checkmark$$

$$\angle BAT = 21^\circ \text{ (alternate segment theorem)} \quad \checkmark$$

$$(f) \quad i. \quad \text{Gradient of line } PQ = \frac{p^2 - q^2}{2p - 2q} = \frac{(p-q)(p+q)}{2(p-q)} = \frac{p+q}{2}$$

Equation of chord PQ :

$$y - p^2 = \frac{p+q}{2}(x - 2p)$$

$$y = \frac{p+q}{2}x - pq \quad \checkmark \text{ (or similar)}$$

ii.

$$\begin{aligned}
 M &= \left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right) \\
 &= \left(p+q, \frac{p^2+q^2}{2} \right) \quad \checkmark
 \end{aligned}$$

iii.

chord passes through $A(0, 2)$

$$\begin{aligned}
 2 &= -pq \\
 -2 &= pq \quad \checkmark \text{ (show that question)}
 \end{aligned}$$

iv. From, 12bii, $x = p + q$ and $y = \frac{p^2+q^2}{2}$

$$\begin{aligned}
 y &= \frac{(p+q)^2 - 2pq}{2} \\
 &= \frac{x^2 - 2(-2)}{2} \\
 y &= \frac{1}{2}x^2 + 2 \quad \checkmark
 \end{aligned}$$

13. (a) i. When $t = \tan \frac{\theta}{2}$, $\sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$

$$\begin{aligned}
 \sin \theta + \cos \theta &= \frac{1}{4} \\
 \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} &= \frac{1}{4} \quad \checkmark \\
 \frac{2t+1-t^2}{1+t^2} &= \frac{1}{4} \\
 8t+4-4t^2 &= 1+t^2 \\
 5t^2-8t-3 &= 0 \quad \checkmark \text{ (show that question)}
 \end{aligned}$$

ii.

$$\begin{aligned}
 5t^2 - 8t - 3 &= 0 \\
 t &= \frac{8 \pm \sqrt{(-8)^2 - 4 \times 5 \times -3}}{2 \times 5} \\
 t &= \frac{4 \pm \sqrt{31}}{5} \quad \checkmark \\
 \text{so } \tan \frac{\theta}{2} &= \frac{4 \pm \sqrt{31}}{5} \\
 \frac{\theta}{2} &\approx 1.08924 \text{ or } \frac{\theta}{2} \approx -0.30384 \\
 \theta &\approx 2.2 \text{ or } \theta \approx -0.6 \quad \checkmark \text{ (both)}
 \end{aligned}$$

(b) i. $\tan \theta = \frac{x}{2}$

$$\text{When } x = \frac{2}{\sqrt{3}}, \theta = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6} \quad \checkmark$$

ii. So $\frac{dx}{dt} = 16$

$$\frac{dx}{d\theta} = 2 \sec^2 \theta \text{ and } \frac{d\theta}{dx} = \frac{1}{2} \cos^2 \theta \quad \checkmark$$

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{d\theta}{dx} \times \frac{dx}{dt} \\ &= \frac{1}{2} \cos^2 \theta \times 16 \\ &= \frac{1}{2} \cos^2 \frac{\pi}{6} \times 16 \\ &= 6 \text{ rad/s} \quad \checkmark \end{aligned}$$

iii. In $\triangle ORQ$, $OR = OQ = 1\text{m}$ (radii) and $\triangle ORQ$ is isosceles so $\angle ORQ = \theta$ (base angles of isosceles triangles are equal) and $\angle ROQ = 180^\circ - 2\theta$ (angles in a triangle).

$$\text{Area } \triangle ORQ = \frac{1}{2} \times 1 \times 1 \times \sin(180^\circ - 2\theta) = \frac{1}{2} \sin 2\theta \quad \checkmark$$

$\angle QOT = 2\theta$ (angle at centre double angle at circumference)

$$\text{Area Sector } OQT = \frac{1}{2} \times 1^2 \times 2\theta = \theta \quad \checkmark$$

Therefore $A = \theta + \frac{1}{2} \sin 2\theta$

iv.

$$\begin{aligned} \frac{dA}{d\theta} &= 1 + \cos 2\theta \quad \checkmark \\ \frac{dA}{dt} &= \frac{dA}{d\theta} \times \frac{d\theta}{dt} \\ &= (1 + \cos 2\theta) \times 6 \\ &= \left(1 + \cos \frac{\pi}{3}\right) \times 6 \\ &= 9 \text{ m}^2/\text{s} \quad \checkmark \end{aligned}$$

(c) i.

$$(a + bx)^n = \sum_{r=0}^n {}^nC_r a^{n-r} (bx)^r$$

$$t_r = {}^nC_r a^{n-r} b^r$$

$$t_r = {}^nC_{r+1} a^{n-r-1} b^{r+1}$$

$$\frac{t_{r+1}}{t_r} = \frac{{}^nC_{r+1} a^{n-r-1} b^{r+1}}{{}^nC_r a^{n-r} b^r} \quad \checkmark$$

$$= \frac{\frac{n!}{(n-r-1)!(r+1)!} a^{n-r-1} b^{r+1}}{\frac{n!}{(n-r)!r!} a^{n-r} b^r}$$

$$= \frac{n!(n-r)!r!}{n!(n-r-1)!(r+1)!} \times \frac{b}{a} \quad \checkmark$$

$$= \frac{n-r}{r+1} \times \frac{b}{a}$$

ii. Equal coefficients when $\frac{t_{r+1}}{t_r} = 1$. $a = 2$, $b = 3$ and $n = 14$, so

$$1 = \frac{n-r}{r+1} \times \frac{b}{a}$$

$$b(n-r) = a(r+1)$$

$$3 \times (14-r) = 2(r+1)$$

$$42 - 3r = 2r + 2$$

$$40 = 5r$$

$$r = 8 \quad \checkmark$$

when $r = 8$

$$\begin{aligned} t_r &= {}^{14}C_8 2^{14-8} 3^8 \\ &= 1\,260\,971\,712 \quad \checkmark \end{aligned}$$

14. (a) When $P(x)$ is divided by $(x+2)(x-5)$ it may have a linear remainder so we can write:

$$P(x) = (x+2)(x-5)Q(x) + Ax + B,$$

where $Q(x)$ is a polynomial and A and B are constants.

$$P(-2) = 6 = -2A + B \quad (1)$$

$$P(5) = -1 = 5A + B \quad (2) \quad \checkmark \quad (\text{both})$$

$$(2) - (1): \quad -7 = 7A \text{ so } A = -1 \text{ and } B = 4$$

So the remainder is $4 - x \quad \checkmark$

(b) i.

$$\begin{array}{ll}
\ddot{x} = 0 & \ddot{y} = -10 \\
\dot{x} = C_1 & \dot{y} = -10t + C_3 \\
\text{when } t = 0, \dot{x} = 15 \cos \alpha & \text{when } t = 0, \dot{y} = 15 \sin \alpha \\
\dot{x} = 15 \cos \alpha & \dot{y} = -10t + 15 \sin \alpha \\
x = 15t \cos \alpha + C_2 & y = -5t^2 + 15t \sin \alpha + C_4 \\
\text{when } t = 0, x = 0 \cos \alpha & \text{when } t = 0, y = 0 \\
x = 15t \cos \alpha & y = -5t^2 + 15t \sin \alpha \quad \checkmark \text{ (both)}
\end{array}$$

ii. Maximum height when $\dot{y} = 0$, so:

$$\begin{array}{l}
0 = -10t + 15 \sin \alpha \\
t = \frac{3}{2} \sin \alpha \quad \checkmark
\end{array}$$

Maximum height at $t = \frac{3}{2} \sin \alpha$ and $y = h$:

$$\begin{array}{l}
h = -5 \left(\frac{3}{2} \sin \alpha \right)^2 + 15 \times \frac{3}{2} \sin \alpha \times \sin \alpha \\
h = -\frac{45}{4} \sin^2 \alpha + \frac{45}{2} \sin^2 \alpha \\
h = \frac{45}{4} \sin^2 \alpha \quad \checkmark \text{ (show that question)}
\end{array}$$

iii. As the motion is symmetrical, it returns back to the initial height at twice the time taken to reach maximum height. Therefore it returns to the initial height at $t = 3 \sin \alpha$. $\checkmark$

When $t = 3 \sin \alpha$:

$$\begin{array}{l}
x = 15 \times 3 \sin \alpha \times \cos \alpha \\
= 45 \sin \alpha \cos \alpha \\
= \frac{45}{2} \sin 2\alpha \quad \checkmark \text{ (show that question)}
\end{array}$$

iv. Taking the point of projection as the origin, the paper bin is at the same height. Therefore the maximum height that is possible is 3m.

From 14bi)

$$\begin{array}{l}
3 = \frac{45}{4} \sin^2 \alpha \\
\frac{4}{15} = \sin^2 \alpha \\
\sin \alpha = \pm \frac{2\sqrt{15}}{15} \\
\text{as the angle of projection is positive, } \sin \alpha = \frac{2\sqrt{15}}{15} \quad \checkmark \text{ (or similar)}
\end{array}$$

the maximum distance is given when $\alpha = \frac{\pi}{4}$ as this α is less than $\frac{\pi}{4}$ it must be the maximum distance possible.

substitute this into the formula in 14bii):

$$x = 45 \sin \alpha \cos \alpha$$

if $\sin \alpha = \frac{2\sqrt{15}}{15}$, $\cos \alpha = \frac{\sqrt{165}}{15}$ using Pythagoras

$$x = 45 \times \frac{2\sqrt{15}}{15} \times \frac{\sqrt{165}}{15}$$

$$x = 6\sqrt{11}\text{m} \quad \checkmark \text{ (show that question)}$$

(c)

Prove true for $n = 0$

$$\begin{aligned} LHS &= \frac{1}{2^0} \tan \left(\frac{x}{2^0} \right) \\ &= \tan x \end{aligned}$$

$$\begin{aligned} RHS &= \frac{1}{2^0} \cot \left(\frac{x}{2^0} \right) - 2 \cot(2x) \\ &= \cot x - 2 \cot 2x \\ &= \frac{1}{\tan x} - \frac{2}{\tan 2x} \\ &= \frac{1}{\tan x} - \frac{2 - 2 \tan^2 x}{\tan 2x} \\ &= \frac{2 - 2 + 2 \tan^2 x}{2 \tan x} \\ &= \tan x \\ &= LHS \quad \checkmark \end{aligned}$$

Assume true for $n = k$

$$\sum_{r=0}^k \frac{1}{2^r} \tan \left(\frac{x}{2^r} \right) = \frac{1}{2^k} \cot \left(\frac{x}{2^k} \right) - 2 \cot(2x) \text{ Prove true for } n = k + 1$$

Required to Prove:

$$\sum_{r=0}^{k+1} \frac{1}{2^r} \tan \left(\frac{x}{2^r} \right) = \frac{1}{2^{k+1}} \cot \left(\frac{x}{2^{k+1}} \right) - 2 \cot(2x)$$

Proof:

$$\begin{aligned}
 LHS &= \sum_{r=0}^{k+1} \frac{1}{2^r} \tan\left(\frac{x}{2^r}\right) \\
 &= \sum_{r=0}^k \frac{1}{2^r} \tan\left(\frac{x}{2^r}\right) + \frac{1}{2^{k+1}} \tan\left(\frac{x}{2^{k+1}}\right) \\
 \text{(using assumption)} \quad &= \frac{1}{2^k} \cot\left(\frac{x}{2^k}\right) - 2 \cot 2x + \frac{1}{2^{k+1}} \tan\left(\frac{x}{2^{k+1}}\right) \quad \checkmark \\
 &= \frac{1}{2^{k+1}} \left(2 \cot\left(\frac{x}{2^k}\right) + \tan\left(\frac{x}{2^{k+1}}\right) \right) - 2 \cot 2x \\
 \text{Let } y &= \frac{x}{2^k} \\
 &= \frac{1}{2^{k+1}} (2 \cot(2y) + \tan y) - 2 \cot 2x \\
 &= \frac{1}{2^{k+1}} \left(\frac{2 \cos(2y)}{\sin(2y)} + \tan y \right) - 2 \cot 2x \\
 &= \frac{1}{2^{k+1}} \left(\frac{2 \cos^2 y - 2 \sin^2 y}{2 \sin y \cos y} + \tan y \right) - 2 \cot 2x \\
 &= \frac{1}{2^{k+1}} (\cot y - \tan y + \tan y) - 2 \cot 2x \\
 &= \frac{1}{2^{k+1}} \cot\left(\frac{x}{2^{k+1}}\right) - 2 \cot 2x \\
 &= RHS \quad \text{Hence true for all } n \geq 0 \text{ by induction} \quad \checkmark
 \end{aligned}$$

(d) Roots of $8x^2 - 5x + a = 0$ are $\sin \theta$ and $\cos 2\theta$

$$\begin{aligned}
 \sin \theta + \cos 2\theta &= \frac{5}{8} & (1) & \quad \text{(from sum of roots)} \\
 \sin \theta \cos 2\theta &= \frac{a}{8} & (2) & \quad \text{(from product of roots)} \quad \checkmark \text{ (both)} \\
 (1) \quad 8 \sin \theta + 8 \cos 2\theta - 5 &= 0 \\
 8 \sin \theta + 8 (1 - 2 \sin^2 \theta) - 5 &= 0 \\
 -16 \sin^2 \theta + 8 \sin \theta + 3 &= 0 \\
 16 \sin^2 \theta - 8 \sin \theta - 3 &= 0 \\
 (4 \sin \theta - 3)(4 \sin \theta + 1) &= 0 \\
 \sin \theta &= \frac{3}{4} \quad \text{or } \sin \theta = -\frac{1}{4} \quad \checkmark \\
 \cos 2\theta &= 1 - 2 \sin^2 \theta \\
 &= -\frac{1}{8} \quad \text{or } = \frac{7}{8} \\
 (2) \quad \sin \theta \cos 2\theta &= \frac{a}{8} \\
 \frac{3}{4} \times -\frac{1}{8} &= \frac{a}{8} \quad \text{or } -\frac{1}{4} \times \frac{7}{8} = \frac{a}{8} \\
 a &= -\frac{3}{4} \quad \text{or } a = -\frac{7}{4} \quad \checkmark
 \end{aligned}$$

End of Solutions